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Topic list for the knowledge base. One line per note; follow the link for the full document.

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AI-ML-Data

Topic	Note	Covers
AI Models	ai-models.md	What a model is, training vs inference, learning paradigms, discriminative vs generative, model families, foundation-model stack, choosing a model
— LLMs	llms.md	Tokens, attention, context window, training stages, decoding controls, model behaviour → pipeline design , pipeline patterns, adaptation strategy, evals, cost & latency

Topic	Note	Covers
Agentic systems	agentic-systems.md	Single source of truth for the Project Mentor guiding agent: the task→capability→tracking→benchmark framework, memory layers (the ten stores, their read/write policies and precedence rules), agent registry (inputs and expected outputs of all 8 model calls), 22-tool catalogue, 18-task inventory with the one-loop fit analysis, debug sheet (tags, root causes, benchmark cases), the issue & fix register, the open-questions log (Q-01...) , the planned-work backlog (P-01...) , and the end-to-end implementation walkthrough

Programming-Tech

Currently filed under `Work-Skills/Programming-Tech/` (these began as work-evidence notes and grew into topic notes).

Topic	Note	Covers
Agentic software development	agentic-development-handbook.md	Claude Code internals, the agent loop, harness engineering, commands vs agents vs skills
Software development	eng-labs-platform.md	Source of truth for the eng-labs monorepo: system-at-a-glance figures, monorepo/workspace/Turborepo architecture, package dependency graph, runtime topology, multi-tenancy & auth model, request lifecycle, full stack, skills & roles

Topic	Note	Covers
Multi-tenancy, AuthN & AuthZ	multi-tenancy.md	Part I — patterns: the five isolation models compared, decision guide, shared-schema data modelling, authentication (the three questions, session vs token, revocation debt, tenant resolution & trust, context switching, cross-app SSO), authorization (the permission-model spectrum, three-gate contract, chokepoints, 401/403/404 leakage, failure modes), the four enforcement layers, the six leak paths, per-model operations, isolation + auth test matrix, 7 interview answers, new-system checklist. Part II — codebase lessons: the auth implementation end-to-end (dual-token path, context precedence, three role systems, SSO bridges, honest gaps, the May 2026 incident), scored self-review, status tracking, module depth, tradeoffs ledger, roadmap, meta-lessons

Topic	Note	Covers
Service architecture	service-architecture.md	Core Domain Service / System-of-Record pattern, Shared-Database anti-pattern, API Gateway, workload-based decomposition, async request-reply, generated clients vs duplicated auth logic, externalized policy services, Cloud Run IAM invoker vs static secrets, security angle of private Cloud Run, multi-region (global LB automatic latency routing vs the DB being the real hard part), frontend multi-zone/path-based routing, Cloud Run Service vs Job vs Function . Case study: eng-labs' <code>apps/api</code> + <code>guide-agent</code> + <code>manuscript-reviewer</code> + <code>interview-service</code>
UX design	user-experience.md	A 13-check UX review list for multi-role products (roles, reach, land, feedback, empty states, scale, handoffs ...) + 2 added checks; case study — Project Mentor track vs team surfaces: orthogonal role modelling, organise-by-job-not-role, group-don't-nest, route-don't-filter, tab-preserving switcher; 12 transferable principles
Infra	gpu-infrastructure-hosting-open-source-models.md	GPU infrastructure for hosting open-source models

Work-Skills

File	Purpose
SKILLS-MATRIX.md	Skills matrix with proficiency levels and backing project evidence
README.md	How the Work-Skills track is organised

Meta

Path	Purpose
CLAUDE.md	Operating instructions for maintaining this knowledge base
_scripts/md-to-pdf.sh	Renders .dot diagrams and converts content .md to PDF